Fair for Whom?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach performance comparisons as evidence for a decision.

Model exact counts, unequal groups and changed metrics, reserving the X/Y sound case for independent evidence.

Keep one habit: Look for correct denominators, shared cases and a purpose-sensitive explanation.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Compute a correct-out-of-assessed fraction and explain its **denominator**

Keep the metric definition visible: this lesson measures **accuracy** among assessed classification outcomes and separately reports unassessed selected cases.

It does not teach that missing system **output**s should always be removed from every real evaluation; an end-to-end task may define **output** failure as a failure. Here, the scoring scope and unavailable records are explicit.

Fractions compare observed proportions, not certainty or individual confidence.

By the end, I can compare group fractions and combine unequal-sized groups correctly.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- The shared A/B card counts from the lesson and two colours of counters for correct and incorrect outcomes
- Paper fraction strips partitioned into 4 and 12 equal parts
- The independent X/Y sound counts in W4 and a place to record missing outcomes separately

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Look for: Name the target collection, selection list and cases actually assessed

Reference example, not the only acceptable wording: The target is all 24 specified cards across four conditions. The sample is the 6 bright-circle cards. Dim circles, bright squares and dim squares are absent.

A frame containing only bright-circle IDs excludes every other condition, regardless of how randomly IDs are drawn from it.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Look for: Apply eligibility requirements before ranking candidates by preferences

Reference example, not the only acceptable wording: J eligible scores 2; K eligible scores 1; L excluded for Spanish; M excluded because unavailable. Recommend J.

Preference matches cannot compensate for failing the English requirement.

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a recorded **prediction** from the **observation** used to check it

Reference example, not the only acceptable wording: **Input** height 6 cm; **prediction** far because 6 is at least 6; observed distance 3 units; observed category near because 3 is below 4; mismatch.

Height is available to the model before the trial, while travel distance is measured afterward to check its **prediction**.

Seven are assessed. Report the one unassessed case separately and explain why its outcome is unavailable; do not silently count it as correct or incorrect.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Missing and Unbalanced Examples

Recommendations

Prediction Is Not Fact

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

Concept to explain

Fairness requires checking outcomes for different users and situations.

Term	Meaning
fairness	reasoned treatment of outcomes across people and situations
subgroup result	performance measured for one relevant group
gap	difference between group outcomes
error cost	harm or burden caused by a mistake

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach denominators, conditions and metric choices

> Model shared-case comparisons without letting overall accuracy stand in for fairness.

Define the assessed denominator and A/B test

For this week, **accuracy** means the number of correctly labelled cases divided by the number of assessed cases.

Every assessed case has a system label and a checked reference label, so its result can be judged correct or incorrect. State the system version, cases and scoring rule with the fraction.

Cases without an assessable outcome are reported separately.

Two fixed shape-recognition versions, A and B, label the same 12 previously unseen picture cards. Eight cards are bright and four are dim; each has a checked circle or square reference label.

Both versions return labels for all 12. A is correct on 8 bright and 1 dim card. B is correct on 6 bright and 2 dim cards.

A therefore has 9 correct out of 12 assessed, written 9/12. Dividing **numerator** and **denominator** by 3 gives 3/4. Three quarters equals 75 out of 100, or 75%. B has 8/12, equal to 2/3.

For this comparison keep B as an exact fraction; a rounded percentage is unnecessary.

Compare equivalent fractions

With the same **denominator** 12, compare the **numerator**s: 9 is greater than 8. A is correct on one more of these cards.

This establishes the ordering on this test under this scoring rule; it does not prove that A will always outperform B on future cards.

To compare fractions with different **denominator**s, rewrite them as **equivalent fractions** with a common **denominator**. For example, 3/4 equals 9/12 and 2/3 equals 8/12.

Changing the written **denominator** does not create new test cases: it only helps compare the proportions.

Model the unequal-group aggregation

Question	Version A	Version B
Bright-card accuracy	8/8 = 1	6/8 = 3/4
Dim-card accuracy	1/4	2/4 = 1/2
Combined accuracy	(8 + 1)/(8 + 4) = 9/12	(6 + 2)/(8 + 4) = 8/12

The bright group has twice as many assessed cards as the dim group. Each individual card counts once in the combined result, so add correct counts and assessed counts.

Do not simply average the two group percentages and call it the all-card **accuracy**: that would give each group equal weight despite their different sizes.

If the intended task especially needs dim-card performance, inspect that result and gather suitable additional **evidence** before choosing. An overall leader can still be weaker in a relevant condition.

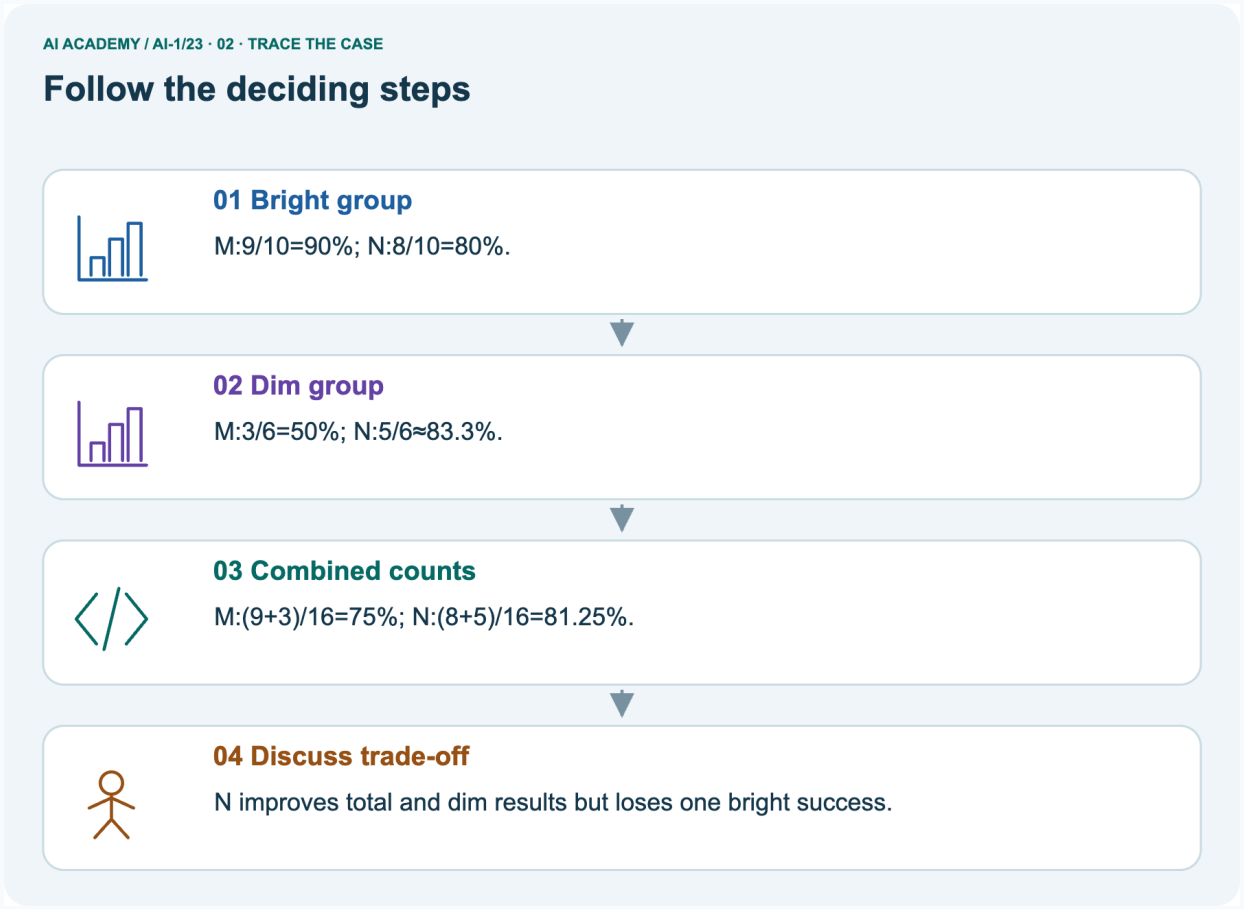
This table alone also does not tell us whether particular mistakes have different consequences. The scoring rule must match the purpose.

Trace the separate badge-reader test

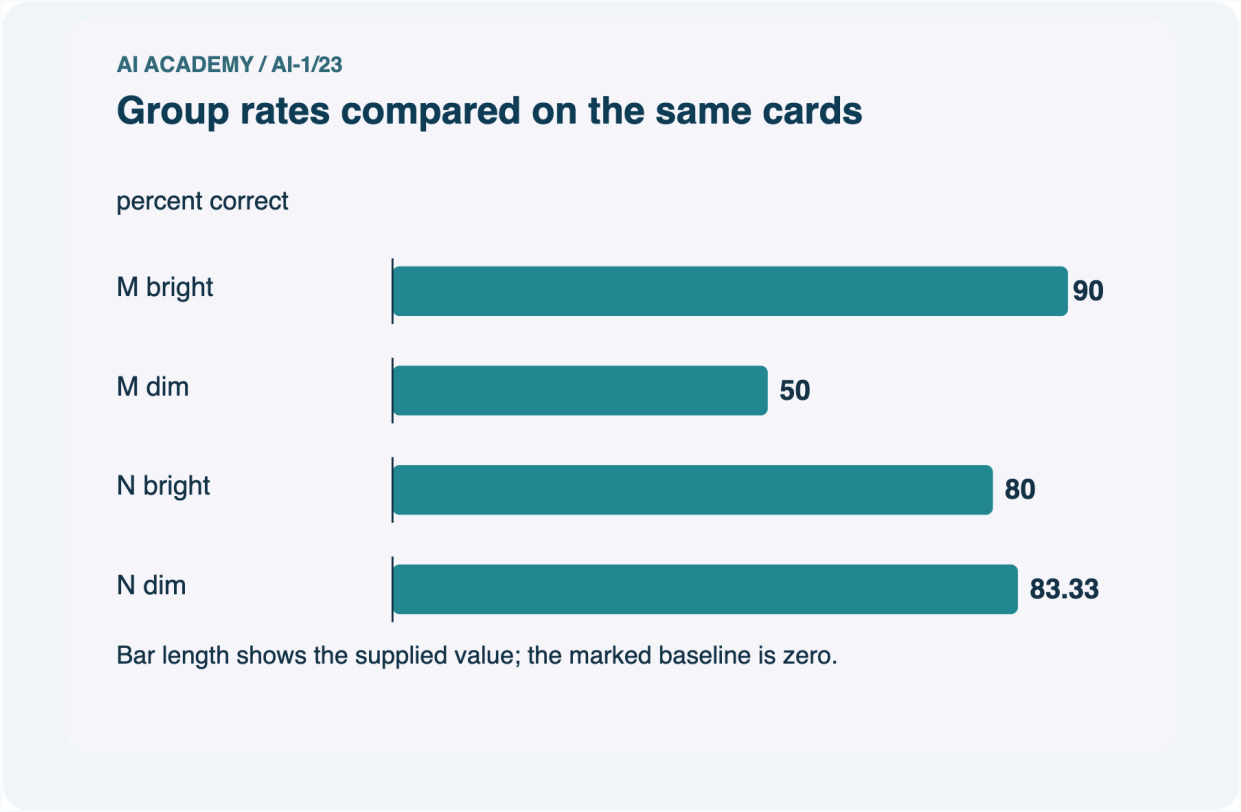
A fictional badge reader is evaluated on bright and dim cards with known reference labels.

Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Bright group	M:9/10=90%; N:8/10=80%.
2. Dim group	M:3/6=50%; N:5/6≈83.3%.
3. Combined counts	M:(9+3)/16=75%; N:(8+5)/16=81.25%.
4. Discuss trade-off	N improves total and dim results but loses one bright success.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

Name the limited evidence

Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

Change the metric explicitly

Change: Give bright and dim groups equal weight as an explicitly different metric.

Prediction: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

Use pictures to support a prediction

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT test cases grouped by relevant user or condition

2. Process — the deciding step

PROCESS compare outcomes and error types by group

3. Output — the result

OUTPUT identify gaps and choose a repair or boundary

CORE IDEA **Fairness** requires checking outcomes for different users and situations.

UNDER THE HOOD **Fairness** is context-specific and cannot be established by overall **accuracy** alone. Compare performance, **error cost**s, access, and remedy across relevant groups while avoiding stereotypes and protecting privacy.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A voice game may work well overall yet miss more commands from quiet speakers than loud speakers.

STEP 1 — OBSERVE test cases grouped by relevant user or condition

STEP 2 — TRACE compare outcomes and error types by group

STEP 3 — CONCLUDE identify gaps and choose a repair or boundary

MODEL REASONING Report each group's success count and error pattern, not only the total. Improve microphone/**process** testing with consented varied voices and keep a non-voice control so access does not depend on one **input** mode.

DO NOT OVERCLAIM **Fairness** work should reduce burdens without turning group labels into stereotypes.

What new **evidence** could change this answer?

Concrete teaching case

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Compare and propose the next investigation.

Exact explanation to model

The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

Explain the added worked example

Key terms

Term	Meaning
Numerator	The count above the fraction line; here, assessed correct cases
Denominator	The count below the fraction line; here, all assessed cases
Equivalent fractions	Different written fractions that represent the same proportion

Term	Meaning
Accuracy	The fraction of assessed classification cases labelled correctly under a stated rule
Aggregate	A result formed by combining the counts of several groups

Model the reasoning aloud

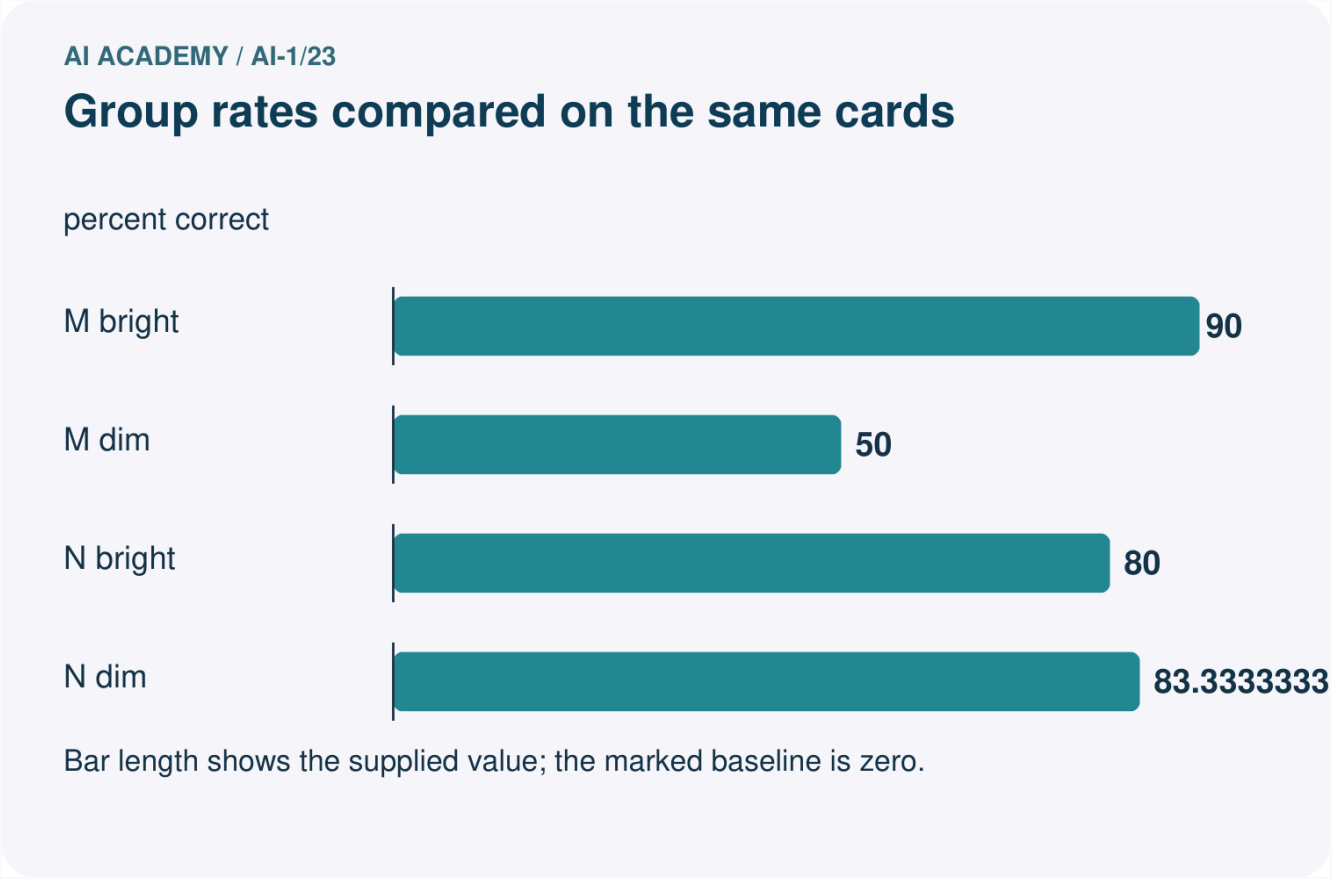
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Fair for Whom?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/23

Fair for Whom?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Bright group

M:9/10=90%; N:8/10=80%.



2. Dim group

M:3/6=50%; N:5/6≈83.3%.



3. Combined counts

M:(9+3)/16=75%; N:(8+5)/16=81.25%.



4. Discuss trade-off

N improves total and dim results but loses one bright success.

CHANGE ONE THING

Give bright and dim groups equal weight as an explicitly different metric.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Compare A and B on the shared 12-card test and on its bright and dim groups. State the leader in each and explain why the overall result does not imply that one version wins in every condition.

Expected reasoning and fading help

Teacher check: Overall A leads 9/12 to B's 8/12. On bright cards A leads 8/8 to 6/8. On dim cards B leads 2/4 to A's 1/4.

The overall count combines unequal groups; A's overall lead does not mean a lead in the dim group or a guarantee for future cards.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Fractions and performance comparisons. Proposed explanation: "Only correct cases belong in the **denominator**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: All assessed cases, including errors, belong in the assessed-**accuracy denominator**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Which group loses performance under N, and can total improvement erase that fact?

Bright drops from 90% to **80%**. No; retain the **subgroup result** alongside the improved total.

Changed-case reasoning: M scores **70%**; N scores about 81.7%. This averages group rates and is not the original 16-card agreement rate.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Write the question to explain: Compare and propose the next investigation.

Reveal after **prediction**s: The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

Use the earlier changed question as optional later practice. Reserve Compare a new shared sound test for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and **denominator**s for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

C has 9/10 correct among assessed outcomes, with 2 unassessed out of 12 selected. A has 9/12 across all 12.

The assessed sets differ, so recover C's lost results or compare all versions on an explicitly documented common assessed subset using case-level records. A's score on that subset cannot be inferred from the totals.

Exit: 6/10 = **60%** across cases; averaging 100% and 50% gives 75% with equal weight to the two groups, a different summary.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve selected versus assessed counts from Week 22.
7 to 20 minutes	Model 9 correct and 3 incorrect counters. Explain denominator 12 and complete W1.
20 to 33 minutes	Use fraction strips to compare 3/4 and 2/3 in twelfths. Trace the diagram.
33 to 45 minutes	Complete W2 and discuss the overall and dim-condition leaders.
Session 2 start 0 to 8 minutes	Retrieve why the incorrect cases stay in the assessed denominator.
8 to 23 minutes	Complete the guided C comparison and W3. Keep unknown cases separate and avoid inventing A's subset score.

Stage	Teacher action
23 to 38 minutes	Collect W4 independently. Require totals, percentages, condition comparisons and a bounded claim.
38 to 45 minutes	Complete W5 and the exit. Use counters to show why combined case counts differ from an equal-group average.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the paper voice-game audit, R=(7+1)/12=8/12 and S=(5+3)/12=8/12. They tie overall, but R leads on loud recordings (7/8 versus 5/8) and S leads on quiet recordings (3/4 versus 1/4). Equal totals do not imply equal condition performance or establish fairness. No sensitive identity inference or actual recording is necessary for this supplied-data activity. Discuss error consequences, relevant follow-up evidence and an alternate non-voice control. Keep Compare a new shared sound test for an individual attempt before feedback.

Run and record

1. Complete the task: Run a voice-game test for quiet and loud speakers.

2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

Only correct cases belong in the **denominator**: All assessed cases, including errors, belong in the assessed-**accuracy denominator**.

Equivalent fractions mean equal test sizes: The proportions match, but 4 assessed cases are not 100 assessed cases; preserve actual counts.

The overall leader wins in every condition: The lesson's dim group and W4's quiet group provide counterexamples.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Compare a new shared sound test for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR A good result for one group means everyone is served well.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Average group percentages to get all-case **accuracy**: Add correct counts and assessed counts when each individual case is weighted equally.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

X has (10 + 2)/(12 + 8) = 12/20 = 3/5 = **60%**. Y has (9 + 5)/20 = 14/20 = 7/10 = **70%**, so Y leads overall. X leads quiet clips 10/12 to 9/12, while Y leads noisy clips 5/8 to 2/8.

These are observed results on this shared test; they do not determine the outcome of the next clip or show that Y is better in every condition.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain both parts of a performance fraction

For version A in the lesson, write correct and assessed counts, simplify the **accuracy** fraction and express it as a percentage. Explain what its **denominator** includes and why 9/9 would be misleading.

A has 9 correct out of 12 assessed: 9/12 = 3/4 = 75%. The **denominator** includes all 12 judged outcomes, including 3 incorrect ones. Using 9/9 would exclude the errors and falsely report perfect assessed **accuracy**.

W2 Compare totals and conditions

Compare A and B on the shared 12-card test and on its bright and dim groups. State the leader in each and explain why the overall result does not imply that one version wins in every condition.

Overall A leads 9/12 to B's 8/12. On bright cards A leads 8/8 to 6/8. On dim cards B leads 2/4 to A's 1/4.

The overall count combines unequal groups; A's overall lead does not mean a lead in the dim group or a guarantee for future cards.

W3 Do not hide unassessed outcomes

C has 9 correct and 1 incorrect assessed outcome, with 2 other selected cases unassessed. Write its assessed **accuracy** and missing count. A reviewer reports 9/12 as assessed **accuracy**; another reports 90% on all 12.

Explain the problem with each label.

C has 9/10 = 90% assessed **accuracy** and 2 unassessed outcomes. The **denominator** 12 includes cases not assessed, so 9/12 is the fraction of selected cases with verified correct outcomes, not assessed **accuracy**.

Reporting 90% on all 12 incorrectly extends the assessed-subset fraction to missing outcomes.

W5 Combine counts rather than group percentages

A separate test has 2 correct out of 2 assessed in group P and 4 correct out of 8 assessed in group Q. Calculate combined **accuracy** as a fraction and percentage. A chart reports 75% by averaging the group percentages.

Explain why that is not the **accuracy** over all assessed cases.

There are 6 correct out of 10 assessed: 6/10 = 3/5 = **60%**. The unweighted mean of 100% and 50% is 75%, but it treats the two differently sized groups equally.

All-case **accuracy** counts each case once, so the larger group contributes 8 of the 10 cases.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

Additional transfer practice question

A helper succeeds on 4 of 4 clear voices and 1 of 4 quiet voices. Who is less well served? What **evidence** should a repair preserve?

Reasoned practice answer

Quiet voices are less well served on these examples. Keep separate counts and test quiet voices after a repair. These small counts flag a concern without proving all users will experience it.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A good result for one group means everyone is served well.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Both are 3/4=75%. Three successes is fewer in count, but not a lower rate. These small samples do not prove equal performance in all quiet and noisy conditions. Collect more varied comparable cases before making a broad claim.

Explanations for the additional workbook examples

W1 Explain both parts of a performance fraction

W2 Compare totals and conditions

W3 Do not hide unassessed outcomes

W4 Compare a new shared sound test

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip’s outcome.

X has (10 + 2)/(12 + 8) = 12/20 = 3/5 = **60%**. Y has (9 + 5)/20 = 14/20 = 7/10 = **70%**, so Y leads overall. X leads quiet clips 10/12 to 9/12, while Y leads noisy clips 5/8 to 2/8.

These are observed results on this shared test; they do not determine the outcome of the next clip or show that Y is better in every condition.

W5 Combine counts rather than group percentages

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use counters and equal-sized fraction strips before percentage notation. Accept exact fractions for group comparisons; rounding is not required.

Extension: describe what per-case records are needed to compare A fairly on C's ten assessed cases, and why the group totals cannot recover them.

W4 requires X 12/20 = **60%**, Y 14/20 = **70%**, Y overall, X quiet, Y noisy and no next-case guarantee.

After support change Y's noisy correct count to 3: Y becomes 12/20 = **60%**, tying X overall while X still leads quiet and Y still leads noisy. Require the changed conclusion with unchanged **denominator**s.

Plan the delayed check

Suggested return points in this level: ai-1/24, ai-1/26, ai-1/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 24: Privacy by Choosing Less. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 24 asks which information a task actually needs and how to reduce unnecessary collection and exposure of personal data.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Compute a correct-out-of-assessed fraction and explain its **denominator**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Compare A and B on the shared 12-card test and on its bright and dim groups.

State the leader in each and explain why the overall result does not imply that one version wins in every condition.

Reasoned practice answer: Overall A leads 9/12 to B's 8/12. On bright cards A leads 8/8 to 6/8. On dim cards B leads 2/4 to A's 1/4.

The overall count combines unequal groups; A's overall lead does not mean a lead in the dim group or a guarantee for future cards.

Check every learner before discussion: Use the worked case for Fractions and performance comparisons. Proposed explanation: "Only correct cases belong in the **denominator**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: All assessed cases, including errors, belong in the assessed-**accuracy denominator**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Give bright and dim groups equal weight as an explicitly different metric. Require a **prediction** and a reason before comparing. Expected result: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

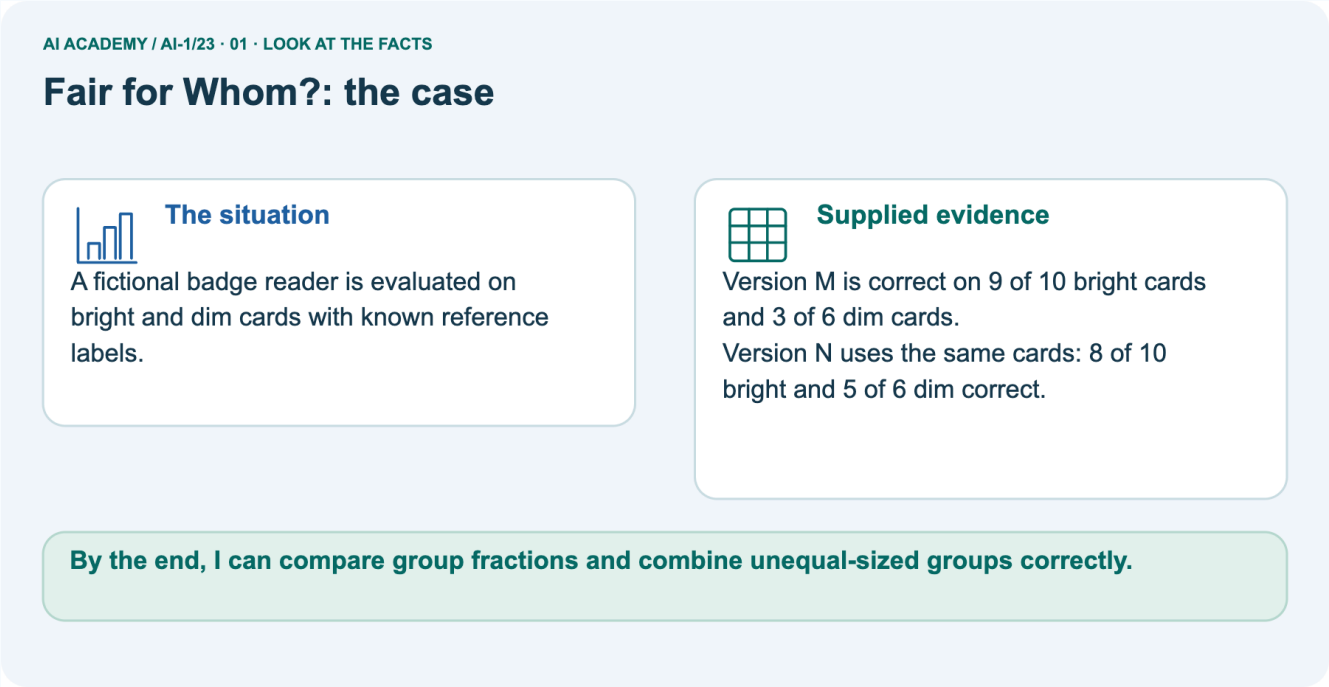
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Fair for Whom?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

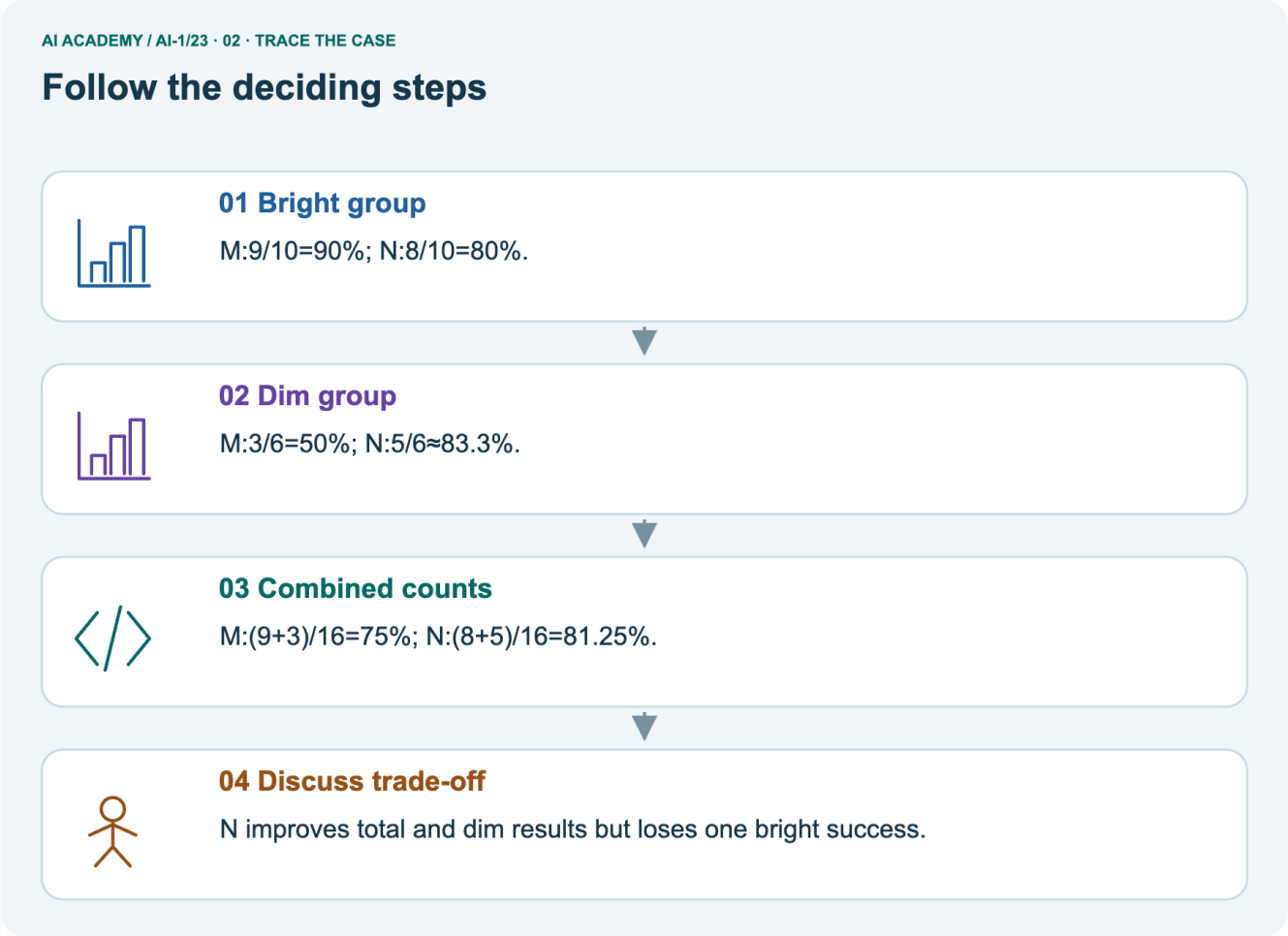
A fictional badge reader is evaluated on bright and dim cards with known reference labels.
Supplied facts: Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



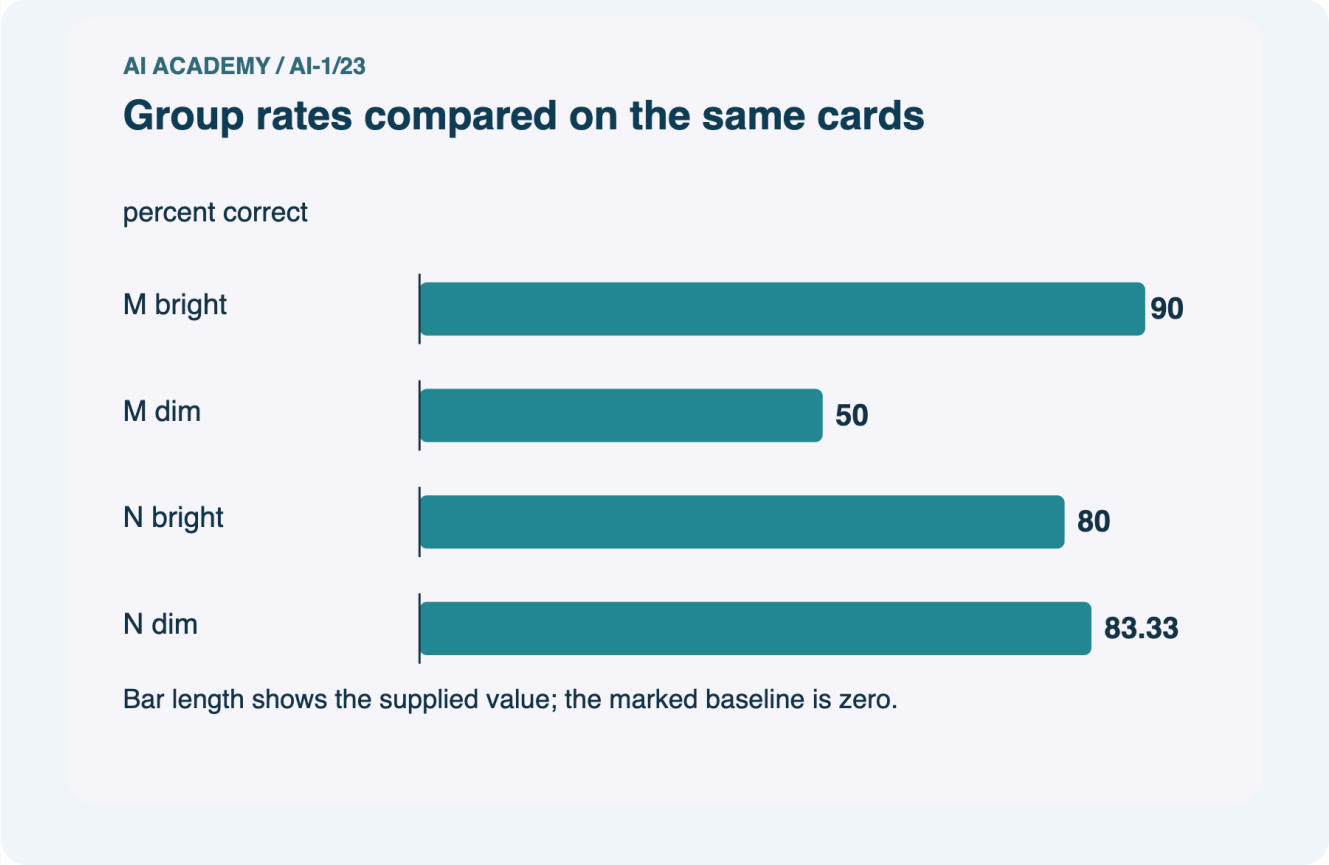
Follow the labelled picture with the matching explanation. Open the vector diagram

Bright group: M:9/10=90%; N:8/10=**80%**. → Dim group: M:3/6=50%; N:5/6≈83.3%. → Combined counts: M:(9+3)/16=75%; N:(8+5)/16=81.25%. → Discuss trade-off: N improves total and dim results but loses one bright success.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Group rates compared on the same cards



Follow the labelled picture with the matching explanation. Open the vector diagram

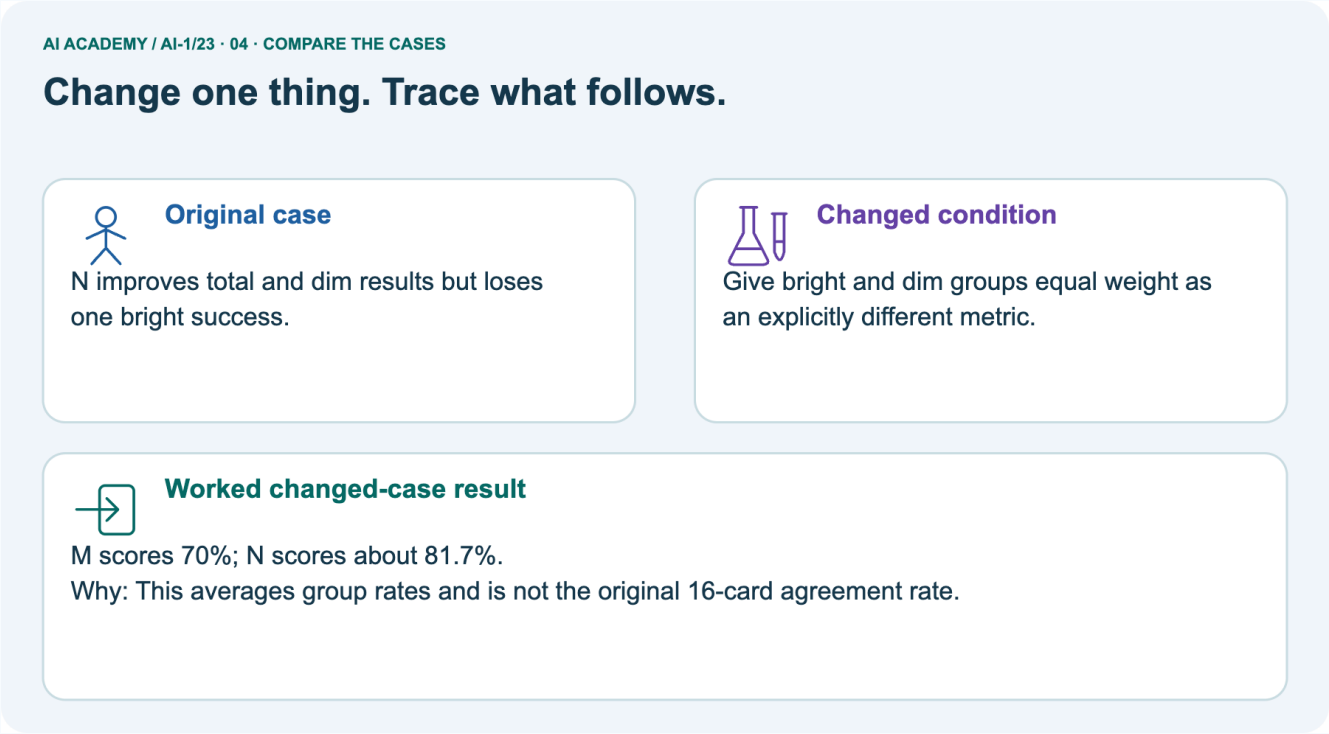
Group rates compared on the same cards. Units: percent correct. M bright: 90. M dim: 50. N bright: 80. N dim: 83.33333333333333.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and **denominator**s before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: N improves total and dim results but loses one bright success. Changed condition: Give bright and dim groups equal weight as an explicitly different metric. Expected result: M scores **70%**; N scores about 81.7%.

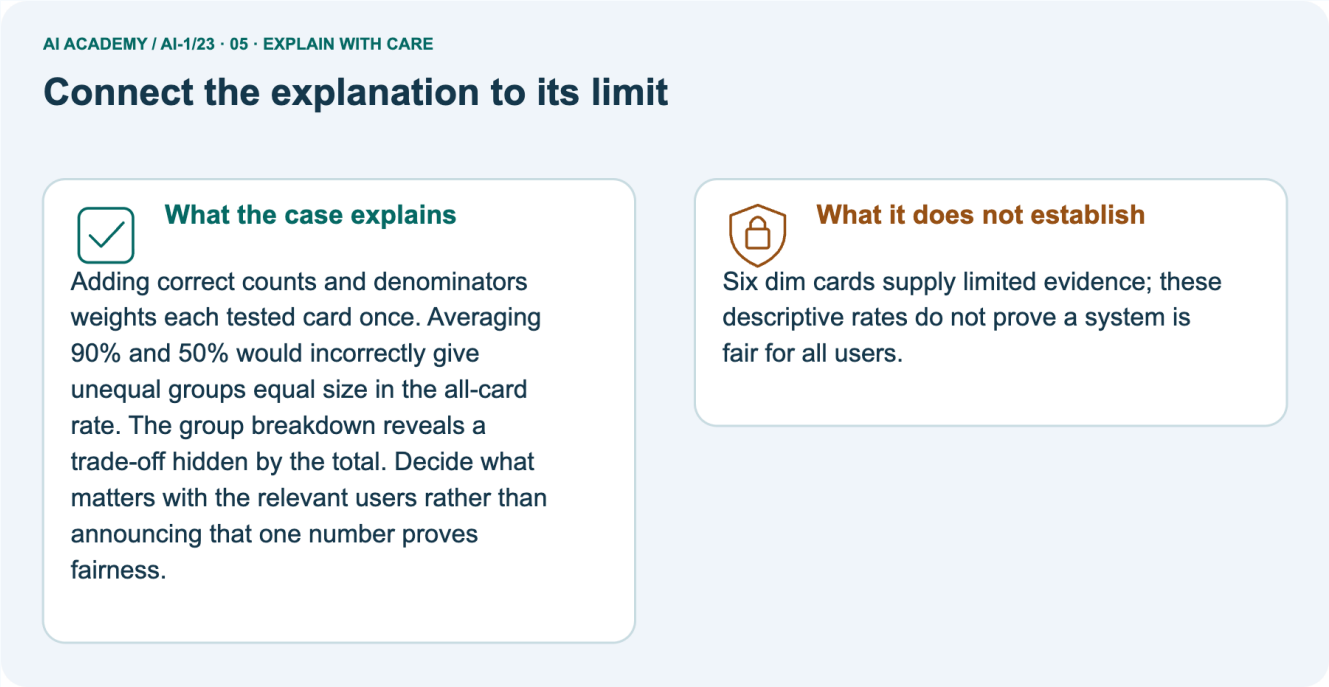
Reason: This averages group rates and is not the original 16-card agreement rate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: This averages group rates and is not the original 16-card agreement rate.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

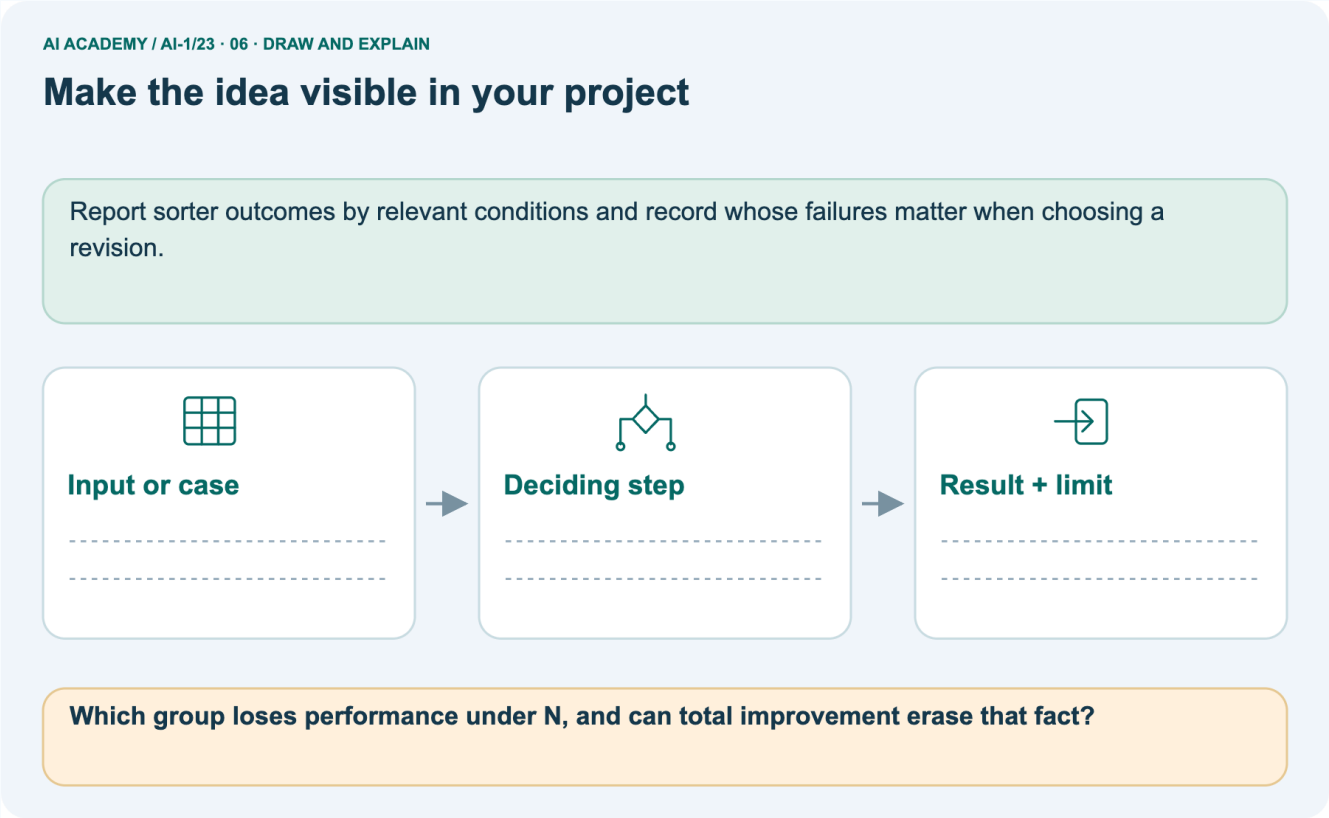
Decide what matters with the relevant users rather than announcing that one number proves **fairness**. Limit: Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision. Draw your **input**, deciding step, result and limit.

Supported question: Which group loses performance under N, and can total improvement erase that fact?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Bright drops from 90% to **80%**. No; retain the **subgroup result** alongside the improved total.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Fair for Whom?: the case

02. Follow the deciding steps

03. Group rates compared on the same cards

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.